

Particulate Matter Monitoring & Health Effects

Issue 2026-08-16 | Entry window 16 August 2026 | 10 records carried, 15 rejected

10RECORDS CARRIED
(15 REJECTED)**6**SUBTOPIC CLUSTERS
REPRESENTED**11**EFFECT ESTIMATES
WITH INTERVALS**5**INSTRUMENTATION /
EXPOSURE RECORDS**2**PREPRINTS
(LABELLED)

Signal of the day

Two very large paediatric studies land on the same day and both report that the constituent, not the mass, carries the effect — and both identify organic matter as the dominant contributor.

Wu et al. (*Ecotoxicol Environ Saf*) run quantile g-computation on **339,435 mother–child pairs** in Liaoning and find prenatal PM_{2.5} total mass at **OR 1.050 (95% CI 1.036–1.064) per IQR** for overweight/obesity by age two, while the constituent mixture reaches **1.063 (1.050–1.076)** with **organic matter contributing 54.1%** of the joint effect, ammonium 34.4% and nitrate 11.5%. Wang et al. (same journal, **449,894 children under five** in China, India and 26 African countries) report cumulative PM_{2.5} at **OR 1.019 (1.009–1.028) in China, 1.009 (1.008–1.009) in India and 1.001 (1.000–1.002) in Africa** for anaemia, with organic carbon and dust the leading components and population attributable fractions of **28.5%, 10.6% and 0.69%** respectively. The two papers disagree on almost everything else — design, outcome, exposure product (Chinese reanalysis vs. MERRA-2), continent — and converge on the same methodological demand: *a mass-only instrument cannot measure the quantity that carries the association*. For a distributed low-cost network this is the recurring problem in its sharpest form, because optical scattering is precisely the technique least able to separate organic matter from ammonium nitrate. Note also that the PAF ordering is inverted relative to the exposure ordering — Africa has the highest anaemia prevalence (64.5%) and the lowest attributable fraction — which is a statement about model-assigned exposure contrast, not about African air.

What changed today

- **Effect modification, not main effect, is where the COVID–PM_{2.5} signal now sits.** Kanno et al. (*JESEE*) find short-term PM_{2.5} alone essentially null for COVID-19 pneumonia (**RR 1.03, 95% CI 0.95–1.12**) and prior myocardial infarction alone at **1.34 (1.14–1.58)**, but the joint contrast at **1.67 (1.41–1.99)**, with upper-quartile PM_{2.5} raising pneumonia risk *only* in those with cardiovascular disease (ATE 0.05, 0.02–0.08). A study powered on the main effect would have concluded nothing.
- **Non-exhaust PM gets a validated inorganic workflow.** Diana et al. (*Talanta*) put 32 WLTP brake-dynamometer PM₁₀ samples through a dual digestion (HNO₃/H₂O₂ and HNO₃/HF/H₃BO₃) validated against NIST SRM 1648a, quantifying **27 elements at >75% recovery**. Low-steel pads on grey cast iron give the highest emission factors and Fe-dominated PM₁₀; non-asbestos organic pads give the lowest and are marked by Ba, Ti, K and Zr. That is a usable source-marker set for apportionment as tailpipe emissions keep falling and brake wear does not.
- **Two independent attacks on the same instrumentation problem: extracting structure from sparse regulatory data.** Perišić et al. (*Sci Total Environ*) cluster SHAP profiles from a single station into ten recurrent meteorology–pollutant settings ($R^2 = 0.913$), showing that identical PM₁₀ concentrations correspond to different physical configurations. Band & Mohammadi (arXiv) pretrain on 10 US stations and adapt to 77 in Taiwan, reaching $R^2 = 0.8739$, **MSE 21.66, MAE 3.14** with an adaptive source encoder. Both are aimed at deployments that cannot afford speciation or dense reference coverage.
- **The day's health records are all exposure-modelled, none exposure-measured.** Every one of the four human-endpoint records assigns exposure from a reanalysis or CMAQ product at ZIP-code, district or 1 km resolution. No record this issue measured particles and outcomes on the same subjects — the third consecutive issue with that property.
- **Weak record worth flagging rather than hiding.** Erickson-DiRenzo et al. (*J Voice*) report paired laryngeal biopsies from **one** participant during a 117 µg/m³ wildfire episode and at <10 µg/m³, with a ≈2-fold MMP-1 increase. $n = 1$, no control for time, phonatory load or infection; carried for the paired-biopsy design, not the result.

Corpus analytics

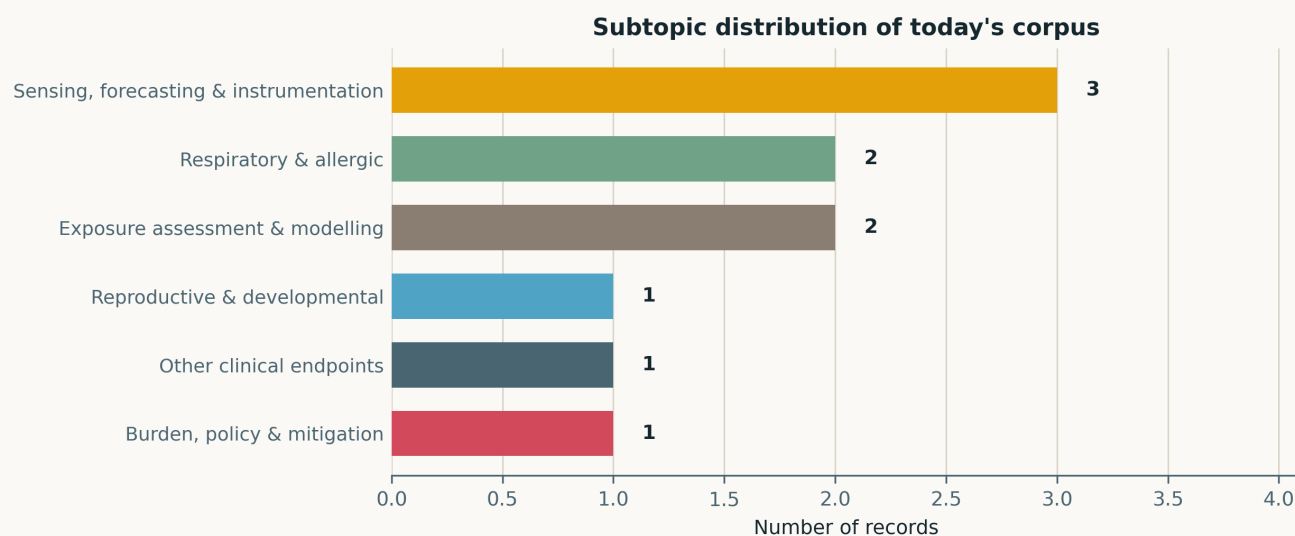


Fig. Subtopic assignment of the 10 carried records, single-label by dominant contribution. Ten records is the smallest issue of the watch to date, tying 5 August; the window is a single Sunday, and PubMed's [EDAT] pipeline deposits almost nothing at the weekend — the connector returned one record for the whole day. Six of ten clusters are occupied. *Sensing, forecasting & instrumentation* leads at 3; *Neuro / mental health*, *Cardiovascular & metabolic*, *Mechanistic toxicology* and *Occupational & indoor* are all empty. *Neuro* is now empty on 11 of 22 issues and remains the watch's least reliable cluster.

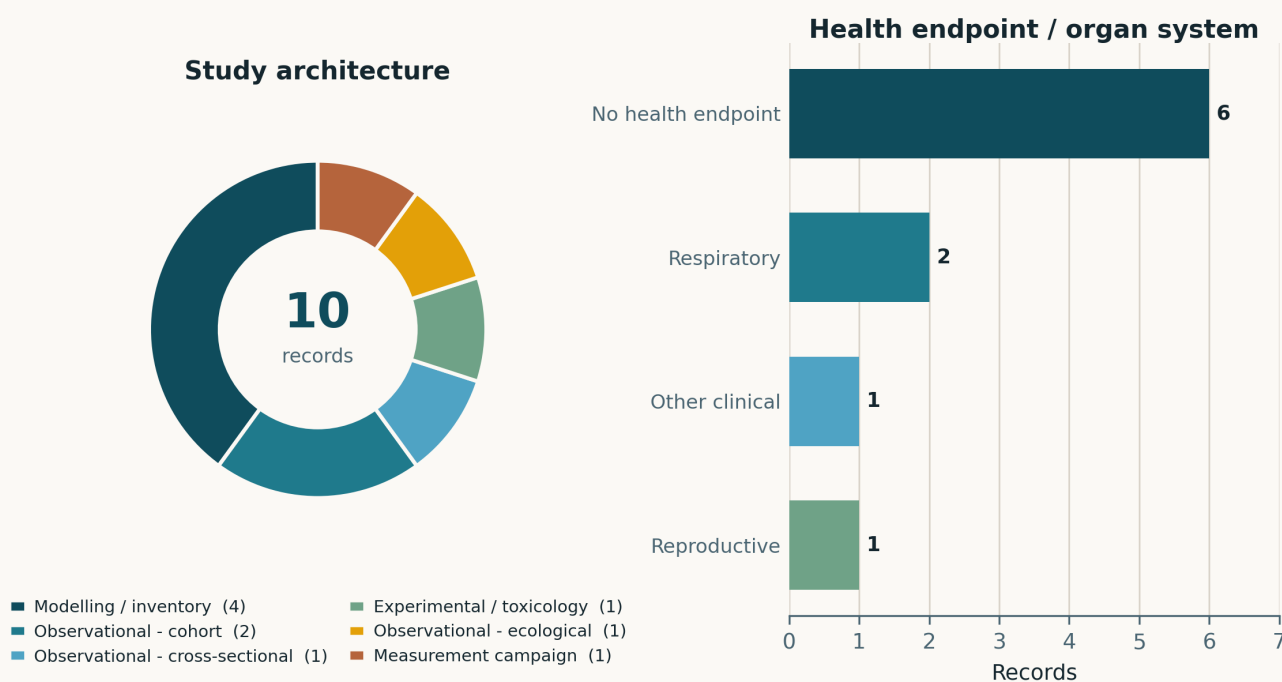


Fig. Left — study architecture. *Modelling / inventory* takes 4 of 10, the largest single block: the XAI station study, the transfer-learning forecaster, the nucleation cluster calculation and the Riyadh satellite preprint. Observational designs split 2 cohort, 1 cross-sectional, 1 ecological; one measurement campaign (the paired laryngeal biopsies) and one bench experiment (the brake dynamometer). No *Trial / intervention* and no *Review / synthesis* record at all. Right — health endpoint. **No health endpoint** is the largest bar at 6 of 10 (60%), within the 41–67% band the last eight issues have occupied once the *None* (...) label family is correctly canonicalised — see the provenance note.

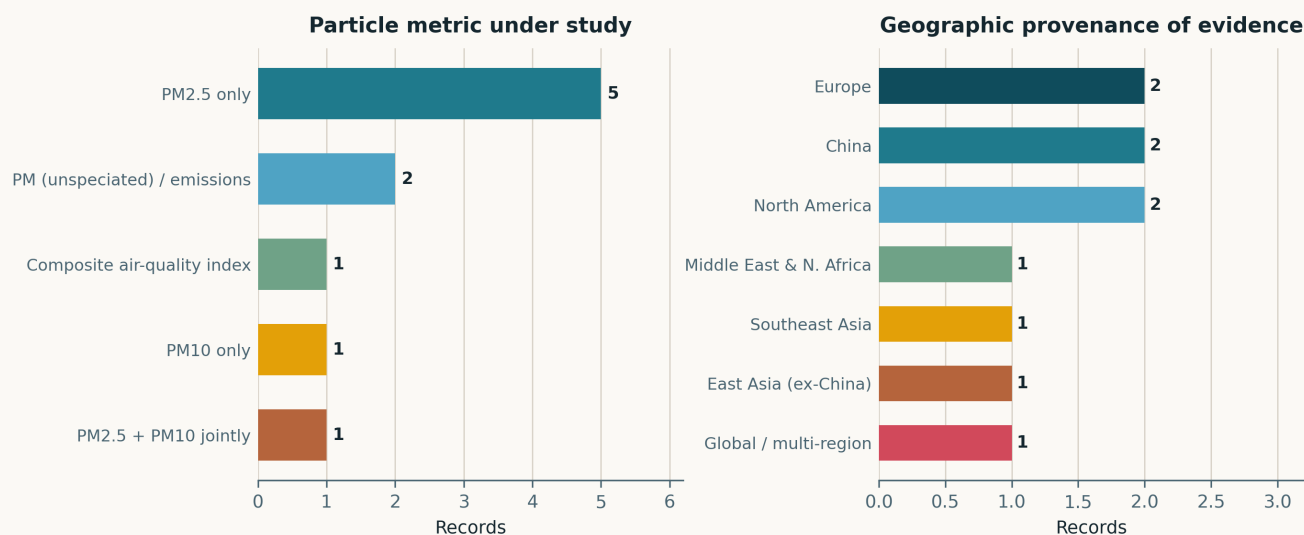


Fig. Left — particle metric. PM_{2.5}-only accounts for 5 records; one is PM₁₀-only (the XAI station study), one is PM_{2.5} and PM₁₀ jointly, and two are unspciated PM or emissions. The single composite-index slice is the Riyadh preprint, which carries *no surface PM product at all* — only TROPOMI trace gases and an Absorbing Aerosol Index; read that slice as an absence. Right — geography. Europe, China and North America hold 2 each; East Asia (ex-China), Southeast Asia, MENA and one multi-region record take the remainder. The multi-region record (Wang et al.) alone spans 28 countries, so the count under-represents the day's actual geographic reach.

Life-course windows addressed today (records may span several stages)

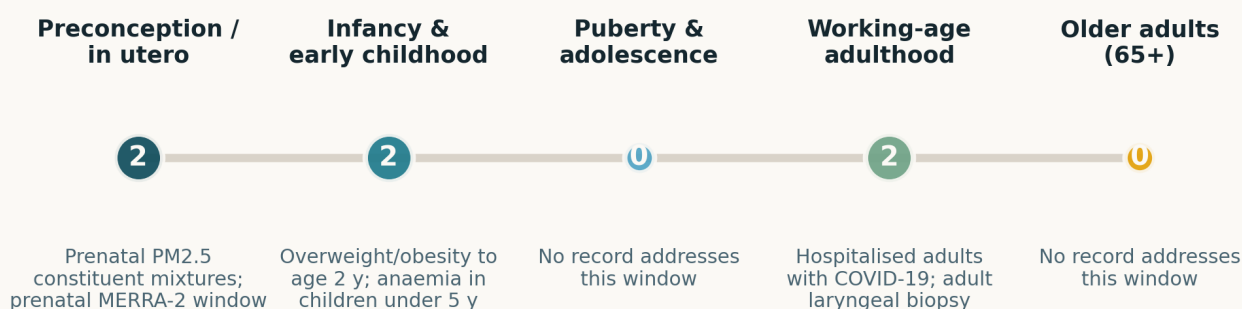


Fig. Life-course windows addressed. Counts are non-exclusive and describe this issue only. Preconception/in-utero and infancy/early childhood hold 2 each — the two large paediatric constituent studies contribute to both, since each assigns a prenatal exposure window to a postnatal outcome. Puberty/adolescence and 65+ are empty. Working age holds 2: the NYC hospitalisation cohort and the single-participant laryngeal study.

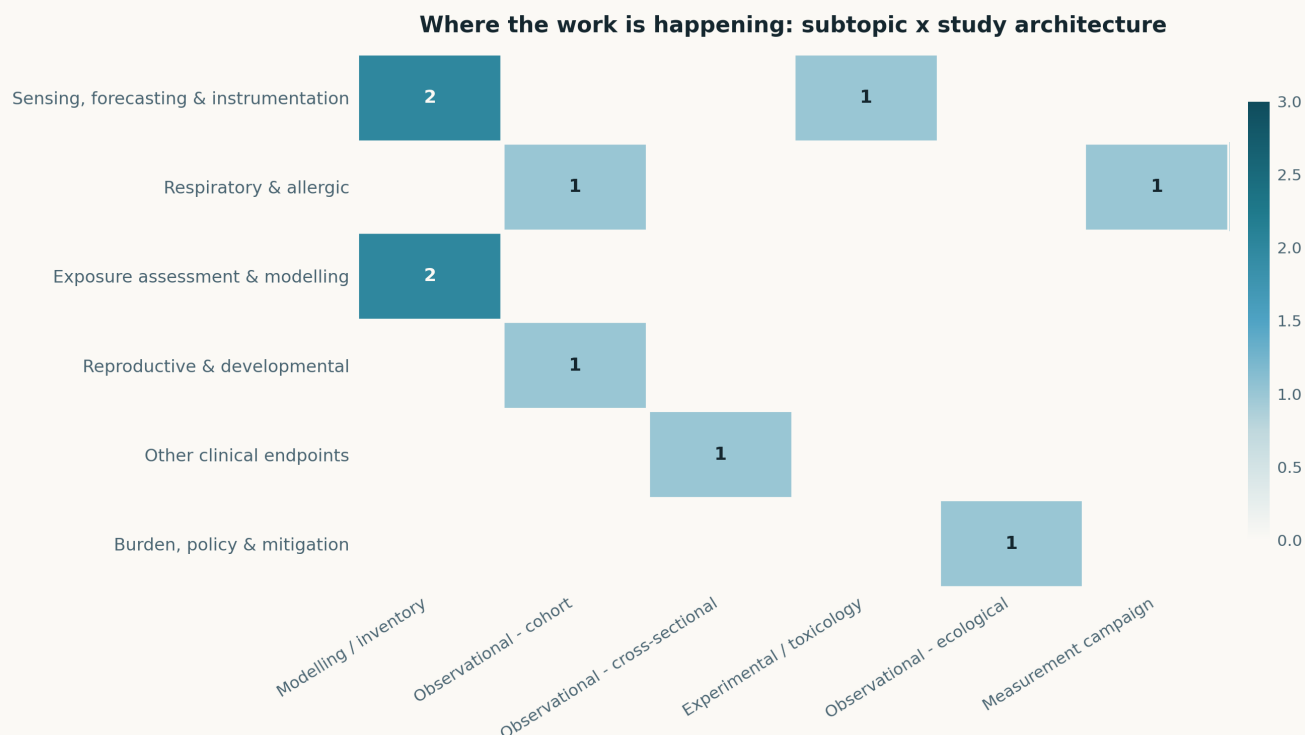


Fig. Subtopic × study-architecture cross-tabulation. Ten records occupy 8 of the 70 available cells and no cell holds more than 2. With a corpus this small the figure carries little information beyond the silo already visible in f2: every *Sensing* and *Exposure* record sits in a *Modelling / inventory*, *Measurement campaign* or *Experimental* column, every human-endpoint record in an observational one, and the two never meet.

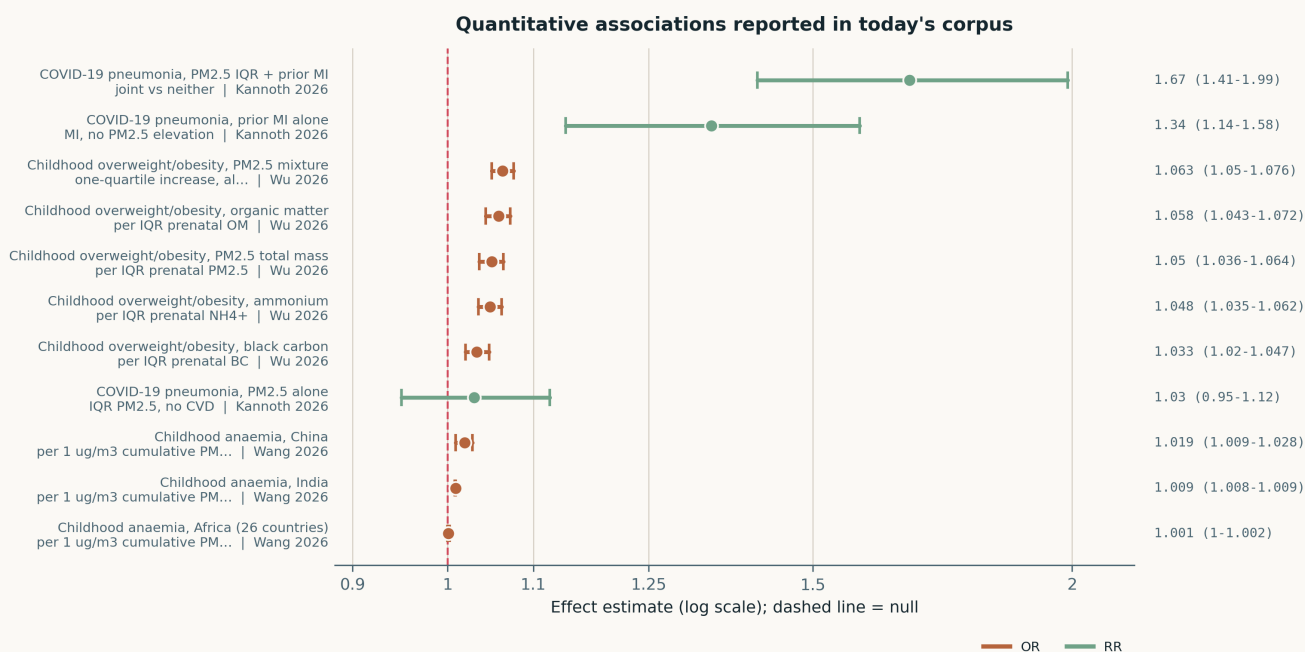


Fig. The 11 interval estimates carried today, on a common logarithmic axis — more estimates than records, because three studies each contribute a family of nested contrasts. **The estimates are not mutually independent.** The three Kannothe contrasts (PM_{2.5} alone, myocardial infarction alone, both jointly) are drawn from the same 15,361-admission cohort and share person-time; the five Wu constituent estimates are IQR-scaled contrasts on correlated constituents of the same mixture in the same 339,435 pairs, so their ordering is informative but their spread is not a between-study spread; the three Wang estimates are region-stratified from one pooled cross-section. Exposure contrasts also differ in kind — Kannothe reports an IQR contrast on a 7-day average, Wu an IQR contrast on a gestational mean, Wang a per-1 $\mu\text{g}/\text{m}^3$ slope on a cumulative window — so interval width across the three families is not comparable and the plot should be read within family, not across it.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Sensing, forecasting & instrumentation

3 records

Perišić et al. 2026 Sci Total Environ

Explainable AI resolves ten recurrent environmental settings behind urban PM₁₀ variability

Four years (2020–2023) of routine regulatory data from a *single* station — PM₁₀, PM_{2.5}, NO₂, SO₂, O₃ and meteorology — fed to ensemble machine-learning models with metaheuristic hyperparameter optimisation, reaching $R^2 = 0.913$, then clustered on SHAP-derived predictor–impact profiles rather than on concentrations. **Ten recurrent settings emerge.** The strongest positive contributions are cold-season accumulation regimes: E0 is frequent and predominantly nocturnal at a mean impact of $50.9 \mu\text{g m}^{-3}$; E7 is rarer but reaches $82.9 \mu\text{g m}^{-3}$ and persists day and night. Warm, well-mixed settings run **26–29 $\mu\text{g m}^{-3}$ below** the model baseline. The substantive claim — that equal concentrations can encode different pollutant–meteorology configurations — is well supported and directly useful where speciation is unaffordable. The limitation is structural and the authors say so: one station, one climate, no external validation, and SHAP attributions are model-conditional, so the “settings” are properties of a fitted ensemble before they are properties of the atmosphere. For distributed low-cost networks the appeal is that the entire input set is what a modest node plus a met sensor already produces.

doi: 10.1016/j.scitotenv.2026.182195

Diana et al. 2026 Talanta

Validated multi-technique workflow for inorganic characterisation of brake-wear PM₁₀

Thirty-two PM₁₀ samples from WLTP-based brake-dynamometer tests on four tribological couples — low-steel (LS) and non-asbestos organic (NAO) pads against grey cast-iron (GCI) or coated discs. A combined acid digestion (HNO₃/H₂O₂ plus HNO₃/HF/H₃BO₃) was validated against **NIST SRM 1648a** and quantified by ICP-OES and ICP-MS, giving **27 elements with recoveries above 75% for most analytes**; FESEM-EDS confirmed the trends at particle scale and PCA/HCA separated the tribological families cleanly. **LS-GCI produced the highest PM₁₀ emission factors and NAO-GCI the lowest**; Fe dominates LS-derived material while Ba, Ti, K and Zr mark NAO. An Mn-based enrichment factor improved discrimination of rotor- versus pad-derived contributions. The value here is a defensible marker set for non-exhaust source apportionment, which matters more each year as tailpipe PM falls. Caveats are the usual dynamometer ones: WLTP braking is not real-world braking duty, only four couples were tested, and no on-road or roadside validation is attempted.

doi: 10.1016/j.talanta.2026.130448

Band & Mohammadi 2026 (preprint) arXiv

Shift-aware dual-encoder transfer learning for short-horizon PM_{2.5} forecasting

A source encoder pretrained on hourly PM_{2.5} from 10 US monitoring locations is fused with a target-specific encoder and adapted to **77 Taiwanese stations** over two years under a chronological train/validation/test split — the correct protocol, and one many comparable papers do not use. Against four baselines the frozen-source dual encoder gives **MSE 21.8960, MAE 3.1597**, $R^2 = 0.8725$ (a 7.1% and 4.1% MSE reduction over two transfer-learning variants); allowing the source encoder to adapt improves this to **MSE 21.6575, MAE 3.1383**, $R^2 = 0.8739$. Ablation shows the Taiwan-specific branch matters most, and SHAP attributes predictions mainly to recent PM_{2.5} and meteorology. The framing — explicit handling of source/target distribution shift to avoid negative transfer — is the right one for anyone standing up a network in a region with no long record. Two reservations: the absolute gain over the better baselines is a few percent of MSE and no significance or repeated-seed variance is reported, and both domains are reference-grade regulatory data, so transfer onto low-cost nodes (a far larger shift) is untested. Preprint, not peer reviewed.

doi: 10.48550/arXiv.2608.14456

Exposure assessment & modelling

2 records

Fan et al. 2026 iScience

Multiple amides synergistically enhance sulfuric acid–ammonia nucleation

Quantum-chemical cluster thermodynamics plus kinetic simulation of new particle formation in binary, ternary and quaternary systems containing sulfuric acid (SA), ammonia, methylacetamide (MAA) and methylpropionamide (MPA). **(SA)₁·(MPA)₁ and (SA)₁·(MAA)₁ are more stable than (SA)₁·(NH₃)₁ and even than (SA)₁·(DMA)₁**, and the two amides together enhance nucleation by **an order of magnitude more than either alone**, most strongly at low SA concentration and low temperature. A displacement hierarchy is identified: MPA replaces NH₃, MAA replaces MPA. Carried because secondary particle number — not mass — is what an optical low-cost sensor is blindest to, and because amide sources (industrial, agricultural, carbon-capture solvent degradation) are rising and largely unmonitored. It is a purely computational result with no chamber or ambient confirmation, the amide concentrations assumed are not tied to measured ambient values, and cluster-level thermodynamics has a poor track record of quantitative transfer to observed formation rates.

doi: 10.1016/j.isci.2026.117013

Alshehri & El-Hamid 2026 (preprint) Research Square

Sentinel-5P and MODIS land-surface coupling of pollutant fields over Riyadh, 2018–2025

Eight years of TROPOMI NO₂, SO₂, CO, O₃ and Absorbing Aerosol Index combined with MODIS land-surface temperature and NDVI and Sentinel-2 land cover in Google Earth Engine, analysed by correlation, PCA and multiple linear regression. NO₂ and CO concentrate over urban and industrial areas, O₃ peaks downwind, LST falls from 43.10 °C (2021) to 37.07 °C (2025) while NDVI rises after 2022. **The regression is weak: R² ranges from 0.124 (O₃) to 0.231 (SO₂)**, and PCA explains 55.8% of variance; the authors correctly conclude that land-surface descriptors explain only a minor share and that emissions and transport dominate. Carried as a negative methodological result for an arid-region setting where ground networks are sparse — and as a caution, because **no surface PM product appears anywhere in the study** despite the framing as air pollution monitoring; AAI is a column absorption index, not a PM concentration. Preprint, not peer reviewed, and the LST trend is reported without any test for its significance against interannual variability.

doi: 10.21203/rs.3.rs-10538140/v1

Respiratory & allergic

2 records

Kanno et al. 2026 J Expo Sci Environ Epidemiol

Short-term PM_{2.5} and COVID-19 pneumonia in New York City, amplified by pre-existing cardiovascular disease

15,361 NYC COVID-19 hospitalisations (March–December 2020) with short-term PM_{2.5} defined as the 7-day residential ZIP-code average from EPA CMAQ, and 15 ICD-coded circulatory conditions as the modifier. Modified Poisson regression for ARDS, pneumonia, ventilation and dialysis; Cox models for fatality and length of stay; RERI for additive interaction; TMLE for average treatment effects. **PM_{2.5} alone was null (RR 1.03, 95% CI 0.95–1.12) and prior myocardial infarction alone 1.34 (1.14–1.58), but jointly 1.67 (1.41–1.99)**; upper-quartile PM_{2.5} raised pneumonia risk only among those with cardiovascular conditions (ATE 0.05, 0.02–0.08). The interaction is the finding and it is the sort a main-effect analysis destroys. Weaknesses are real: ZIP-code CMAQ is a coarse surrogate for personal exposure and its error is spatially structured with deprivation, the effect appears only from July onward with no mechanism offered for the seasonal split, and the cohort is hospitalised patients only, so selection on admission is unaddressed. For network design the implication is concrete — the susceptible stratum is identifiable in advance, which is an argument for siting density by cardiovascular prevalence rather than by population alone.

doi: 10.1038/s41370-026-00957-5

Erickson-DiRenzo et al. 2026 J Voice

Wildfire PM_{2.5} and the human larynx: paired in-event and clean-air assessment in one adult

A single adult assessed twice — during a wildfire episode with PM_{2.5} reaching **117 $\mu\text{g m}^{-3}$** and again at **<10 $\mu\text{g m}^{-3}$** — with acoustic and aerodynamic voice measures, laryngeal endoscopy with stroboscopy, and *ventricular fold biopsies at both time points* analysed by qPCR for extracellular matrix and inflammatory markers. Speaking fundamental frequency fell relative to baseline, endoscopy showed visibly increased laryngeal mucus, and **MMP-1 expression roughly doubled**; other ECM and inflammatory genes moved inconsistently. $n = 1$. There is no control participant, no adjustment for phonatory load, hydration, intercurrent infection or the ordinary diurnal variation in fundamental frequency, and a two-point comparison cannot separate exposure from time. The reason to carry it is the design, not the result: paired human airway tissue across a natural exposure contrast is rare, and it establishes feasibility for a properly powered version. Treat the MMP-1 figure as a hypothesis.

doi: 10.1016/j.jvoice.2026.07.048

Reproductive & developmental

1 record

Wu et al. 2026 Ecotoxicol Environ Saf

Prenatal PM_{2.5} constituents and early childhood obesity: organic matter dominates the mixture

339,435 mother–child pairs in Liaoning Province with prenatal PM_{2.5} and five constituents (organic matter, black carbon, ammonium, nitrate, sulfate) assigned from high-resolution atmospheric reanalysis, and overweight/obesity ascertained across the first two years. Single-pollutant models give PM_{2.5} **mass OR 1.050 (95% CI 1.036–1.064) per IQR**, with OM 1.058 (1.043–1.072), NH_4^+ 1.048 (1.035–1.062), NO_3^- 1.048 (1.033–1.064) and BC 1.033 (1.020–1.047); sulfate is null. Quantile g-computation puts the mixture at **1.063 (1.050–1.076)** per quartile increase with OM **contributing 54.1%**, NH_4^+ 34.4% and NO_3^- 11.5%. The sample size makes the intervals tight, but tight intervals on reanalysis-assigned exposure are a precision claim about the model, not the exposure: constituent fields are far more uncertain than total mass and their errors are correlated with each other, which is exactly the condition under which QGC weights become unstable. Residential address at delivery stands in for the whole gestation, and no gestational windows are resolved. The direction is credible; the 54% attribution to OM is not something the exposure product can really support.

doi: 10.1016/j.ecoenv.2026.120670

Burden, policy & mitigation

1 record

Alamsyah et al. 2026 Environ Sci Pollut Res Int

Industrial intensity and traffic congestion propagate air-pollution health vulnerability across district borders in Java

A health vulnerability index (HVI) for Java built by PCA over satellite pollutant fields, official statistics and OpenStreetMap congestion data, then regressed on industrial activity intensity and congestion in a spatial Durbin model to separate direct from spillover effects. **HVI ranges 29.2–62.3 on a 0–100 scale**, highest in West Java, Jakarta and Banten and lowest in Central Java, Yogyakarta and East Java; the exposure dimension varies most (13.18) against adaptive capacity (9.40) and sensitivity (5.17), and both industrial intensity and congestion act through **direct and indirect (spillover) channels**. The spillover result is the useful one — it says district-level mitigation will leak, which is a real argument for regional rather than municipal air management. But the index is a construct: PCA weights are data-driven rather than health-anchored, no health outcome is used to validate the HVI against observed morbidity, and OpenStreetMap congestion is an availability-biased proxy that is denser precisely in the wealthier districts. Uncertainty and sensitivity analysis is reported, which is more than most index papers do.

doi: 10.1007/s11356-026-38148-2

Other clinical endpoints

1 record

Wang et al. 2026 Ecotoxicol Environ Saf

Early-life PM_{2.5} mass and components and childhood anaemia across China, India and Africa

Cross-sectional analysis of **449,894 children under five** from China, India and 26 African countries, with prenatal, postnatal and cumulative PM_{2.5} and five components (black carbon, organic carbon, sulfate, dust, sea salt) from NASA MERRA-2 reanalysis. Anaemia prevalence is **26.9% in China, 62.6% in India, 64.5% in Africa**. Per 1 $\mu\text{g m}^{-3}$ of cumulative PM_{2.5}: **OR 1.019 (95% CI 1.009–1.028) in China, 1.009 (1.008–1.009) in India, 1.001 (1.000–1.002) in Africa**, with organic carbon and dust positive in all three. Population attributable fractions are **28.46%, 10.64% and 0.69%**. The regional ordering of effect size and PAF runs *opposite* to the ordering of anaemia burden, and the paper does not resolve this: the most economical explanation is that MERRA-2 at $0.5^\circ \times 0.625^\circ$ has little within-country exposure contrast in Africa and correspondingly attenuated coefficients, not that African children are less susceptible. Cross-sectional design cannot order exposure and outcome, and anaemia in these settings is dominated by iron deficiency, malaria and haemoglobinopathy — confounders that co-vary strongly with dust exposure. Read the component ranking, not the PAFs.

doi: 10.1016/j.ecoenv.2026.120658

Corpus register

First author	Focus	Journal	Design	Metric	Region
Reproductive & developmental					
Wu et al.	Prenatal PM _{2.5} and its chemical constituents and early childhood overweight/obesity: organic matter dominates the mixture effect	<i>Ecotoxicol Environ Saf</i>	Prospective cohort (mixtures)	Gestational PM _{2.5} mass + OM, BC, NH ₄ ⁺ , NO ₃ ⁻ , SO ₄ ²⁻ (reanalysis)	China
Respiratory & allergic					
Erickson-DiRenzo et al.	Wildfire PM _{2.5} and the human larynx: paired in-event and clean-air voice, endoscopy and ventricular-fold biopsy in one adult	<i>J Voice</i>	Repeated-measures biomarker	Wildfire episode PM _{2.5} up to 117 µg/m ³ vs <10 µg/m ³ baseline	USA
Kannoth et al.	Short-term PM _{2.5} and COVID-19 pneumonia in New York City: risk amplified by pre-existing cardiovascular disease	<i>J Expo Sci Environ Epidemiol</i>	Retrospective cohort	7-day residential ZIP-code CMAQ PM _{2.5} prior to admission	USA
Sensing, forecasting & instrumentation					
Band & Mohammadi 2026	Shift-aware dual-encoder transfer learning for short-horizon PM _{2.5} forecasting in data-limited target domains	<i>arXiv (preprint)</i>	Deep-learning model	Hourly regulatory PM _{2.5} , 10 US source sites -> 77 Taiwan target stations	Taiwan
Diana et al.	Validated multi-technique workflow for inorganic characterisation of brake-wear PM ₁₀ and source-marker identification	<i>Talanta</i>	Bench / device experiment	WLTP brake-dynamometer PM ₁₀ , low-steel and NAO pads on cast-iron/coated discs	Italy
Perisic et al.	Explainable AI resolves ten recurrent environmental settings behind urban PM ₁₀ variability at a single regulatory station	<i>Sci Total Environ</i>	Algorithm development	Regulatory PM ₁₀ /PM _{2.5} + NO ₂ , SO ₂ , O ₃ , meteorology, 2020-2023	Europe
Exposure assessment & modelling					
Alshehri & El-Hamid 2026	Sentinel-5P and MODIS land-surface coupling of pollutant fields over Riyadh, 2018-2025	<i>Research Square (preprint)</i>	Spatial analysis / GIS	TROPOMI NO ₂ /SO ₂ /CO/O ₃ + Absorbing Aerosol Index; MODIS LST/NDVI	Middle East & N. Africa
Fan et al.	Multiple amides synergistically enhance sulfuric acid-ammonia nucleation by an order of magnitude over single-base systems	<i>iScience</i>	Chemical transport modelling	Modelled SA-NH ₃ -MPA/MAA cluster formation (new particle formation)	China
Burden, policy & mitigation					
Alamsyah et al.	Industrial intensity and traffic congestion propagate air-pollution health vulnerability across district borders in Java	<i>Environ Sci Pollut Res Int</i>	Ecological panel	Satellite pollutant fields + OpenStreetMap congestion, district level	Indonesia
Other clinical endpoints					
Wang et al.	Early-life PM _{2.5} mass and components and childhood anaemia across China, India and 26 African countries	<i>Ecotoxicol Environ Saf</i>	Cross-sectional (multistage)	MERRA-2 prenatal/postnatal/cumulative PM _{2.5} + BC, OC, sulfate, dust, sea salt	Multi-country

Provenance and method

Window. Entry window **2026-08-16 to 2026-08-16**, contiguous with the 15 August issue; no gap. Every record was checked against `state/seen.json` (`pmid` → lowercased `doi` → title signature) before summarising; **11 Europe PMC hits were dropped as already-seen duplicates**.

Harvester defect from 15 August: repaired. The Europe PMC leg in `harvest.py` was windowing on `FIRST_PDATE`, which the source-status notes had already recorded as non-functional for recent windows — it returned 0 for both the 15 and 16 August windows. Switched to `CREATION_DATE` and given cursor-mark pagination to 600 records; the committed leg now returns **20** for this window unaided, and every record in this issue except one came through it.

Per-leg yield. A single Sunday window is close to the floor of what this watch can see. PubMed connector, two separate queries on the health and instrumentation axes: **1 hit total**, the same record on both axes, already present in the Europe PMC set. Local harvester `pubmed_health` 0 and `pubmed_sensing` 0 — PubMed's [EDAT] pipeline deposits almost nothing at the weekend, which is a property of the source and not a failure. Europe PMC primary query 20 hits, 9 new after deduplication, **6 carried**; a second broader Europe PMC sweep on air-quality and aerosol terms returned 36, 15 further new, **3 carried**. arXiv 145 kB with one in-scope entry dated 14 August that the 15 August sweep did not surface, **1 carried**. Crossref by ISSN returned 0 across 18 journals. **OpenAlex HTTP 429, 14th consecutive** — the leg is dead. **Consensus: 8th consecutive zero-carry**. All three returned records were in scope and useful (AutoML indoor calibration, multi-city bi-pollutant LCS calibration, global PurpleAir transferability) but their Crossref created dates are 2026-01-23, 2026-01-20 and 2026-06-15. It ranks by relevance, not recency, and should be treated as a weekly/monthly gap-filler only. ClinicalTrials.gov: no PM or air-pollution registration created or updated inside the window; `trials.json` unchanged.

Second defect repaired: endpoint canonicalisation. `ENDPOINT_CANON` mapped `None` (`monitoring`) but not the rest of the `None` (...) family — `None` (`exposure`), `None` (`instrument`), `None` (`emissions`), `None` (`algorithm`), `None` (`modelling`), `None` (`aerosol chemistry`) and singular variants, **35 records across the store**. All mean the same thing, so `f2b` understated its own largest bar; the 9, 10 and 11 August issues rendered a literal 0 for a category that in fact held 8, 9 and 15 records. `canon_ep` now falls back on a prefix rule, and the corrected series for 7–16 August reads 54, 64, 67, 56, 58, 46, 41, 43, 43, 60%. **The three affected daily PDFs are stale on that panel** and are left as shipped; the store and all future rollups are correct.

Rejections (15). Six keyword collisions where `aerosol` or `exposure` matched an unrelated field (freeze-spray insecticide delivery, *Francisella* aerosol-challenge vaccinology, inhaled siRNA therapeutics, PPE-adherence auditing, lanthanum sediment geochemistry, aqueous biochar sorption); four gas-phase-only records with no particulate measurement (formaldehyde catalysis, formaldehyde occupational olfaction, MgAl-LDH NO₂ sorption, ozone melanogenesis); a meteorology-only allergic-rhinitis DLNM carrying pollution as an adjustment covariate; a therapeutics review naming PM only as a background trigger; two correction/corrigendum notices; and one journal news item reporting a study already carried on 27 July. All logged with reasons to `state/rejected.jsonl`.

Two records are preprints (Band & Mohammadi, arXiv; Alshehri & El-Hamid, Research Square) and are labelled as such in the register and the digest. No record this issue is metadata-only.

Labels. Subtopic, design, particle-metric and geography labels were drawn only from vocabularies already mapped in `plots.py` and `mktable.py`; no new leaf label was written this issue and the unmapped-label diagnostics printed nothing. Assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. `check_dois.py` returned **0 fail, 4 warn** across all 10 records; all four warnings are title-overlap heuristics firing on editorial focus lines (three) or on arXiv's absence from Crossref (one), and each was confirmed by eye against the registered record.

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